

GEELON SO

Ph.D. Candidate, University of California, San Diego
Department of Computer Science and Engineering

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RESEARCH INTERESTS

Machine learning theory, sequential decision making, stochastic analysis, optimization, game theory

EDUCATION

University of California, San Diego
Ph.D. Candidate, Computer Science
Advisors: *Sanjoy Dasgupta & Yian Ma*

La Jolla, CA
Sep 2019—present

Columbia University
M.S. Computer Science
Advisor: *Daniel Hsu*

New York, NY
May 2019

The University of Chicago
B.S. Mathematics with Honors
Advisor: *Stuart Kurtz*

Chicago, IL
Jun 2017

PUBLICATIONS¹

Peer-reviewed

Online consistency of the nearest neighbor rule.

Sanjoy Dasgupta and Geelon So. NeurIPS 2024.

Metric learning from limited pairwise preference comparisons.

Zhi Wang, Geelon So, and Ramya Korlakai Vinayak. UAI 2024.

Convergence of online k -means.

Sanjoy Dasgupta, Gaurav Mahajan, and Geelon So. AISTATS 2022.

Pre-prints/under review

Mixed Nash equilibria are uniformly stable if and only if they are collectively rational.

Yi-An Ma and Geelon So. Working manuscript, 2025.

On the sample complexity of semi-supervised multi-objective learning.

Tobias Wegel, Geelon So, Junhyung Park, and Fanny Yang. Under review, 2025.

Online consistency of k_n -nearest neighbor.

Robi Bhattacharjee, Geelon So, and Sanjoy Dasgupta. Under review, 2025.

Learning halfspaces without synthesized data.

Hadley Black, Kasper Green Larsen, Arya Mazumdar, Barna Saha, Geelon So. Under review, 2025.

Preference optimization on Pareto sets: On a theory of multi-objective optimization.

Abhishek Roy*, Geelon So*, and Yi-An Ma. Under review, 2025.

¹Authors in alphabetical order except when they aren't. Equal contribution indicated by an asterisk.

Technical reports

Active learning with noise.

Geelon So. Master's thesis, Columbia University, 2019.

Quantum computing: efficient prime factorization.

Geelon So. REU report, UChicago, 2015.

Dynamical systems: symbolic dynamics.

Geelon So. REU report, UChicago, 2014.

EXPERIENCE

Simons Institute for the Theory of Computing

Berkeley, CA

Visiting graduate student

Sep 2024—Dec 2024

- Visitor for the Special Year on Large Language Models and Transformers program.

Amazon — Supply Chain Optimization Technologies (SCOT) team

Bellevue, WA

Applied Science Intern

Jun 2024—Sep 2024

- Framed and led the first ML augmentation of SCOT's LITMUS fulfillment simulator, replacing purely mechanistic rules with a data-driven generative component for non-deterministic events (e.g., SLAM times).
- Designed a scalable generative modeling pipeline in PyTorch to simulate correlated package trajectories in parallel, matching aggregate behavior, and not just per-package realism.
- Shipped an MLP (minimum lovable product) integrated with LITMUS; on JFK8 (NYC) validation, improved a key accuracy metric by >15% vs. the prior baseline (team typically targets basis-point gains).
- Operationalized an open-ended problem and delivered production-ready package and docs, enabling continued iteration by a non-ML team.

Seekr (news search engine company)

Carlsbad, CA

Machine learning research and engineering intern

Jun 2022—Jun 2023

- Spearheaded the overhaul of the trending news labeling process; developed a human-in-the-loop system to enable rapid experimentation with diverse rules- and ML-methods for keyword extraction and labeling.
- Engineered a data exploration tool to help developers gain intuition for the underlying news/text data; created modular system, streamlining new and iterative refinement of label extraction methods.
- Built a user-friendly GUI for human evaluation of label quality.
- The incorporation of the redesign resulted in significantly more expressive and meaningful topic labels.

Home Partners of America (REIT company)

Chicago, IL

Data science and engineering intern

Jun 2019—Aug 2019

- Identified, planned and executed a high-impact project given limited resource and time constraints.
- Streamlined ETL for the analytics team by providing API to automatically join, transform and aggregate data, while performing data quality checks with statistical guarantees; eliminated need to understand how the history of business decisions affects the correct way to join tables.
- Designed backend to API to be easily adaptable to future changes in data collection/database.

AumiPhyte Health (healthcare startup)

New York, NY

Machine learning consultant

Feb 2018—Mar 2019

- Designed user-centric system and tools to analyze and process medical texts, presenting recommendations in a whitepaper that detailed motivation, implementation, rationale, and limitations.
- Implemented rules- and machine-learning based text processing methods using python and spaCy to generate label data for small set of medical texts; estimated 0.9 accuracy with 0.9 confidence.

- Developed GUI utilizing active learning techniques to help user rapidly label data.

Options for Youth/University of Chicago (non-profit consulting course)

Chicago, IL

Non-profit consultant

Mar 2016—Jun 2016

- Conducted quantitative/qualitative impact assessment for a local nonprofit by (i) researching and building a model to estimate the return on investment and (ii) interviewing stakeholders and recording impact; provided actionable best practices for using impact assessment to target donors.

UChicago Math REU (research experience for undergrads)

Chicago, IL

Undergraduate researcher

Summers 2014, 2015, 2016

- Studied foundations of mathematics, intuitionistic type theory and category theory; advised by Stuart Kurtz. Studied dynamical systems and quantum computing; advised by Clark Butler, Tori Akin, and Peter May.

Chicago Summer School in Mathematics (analysis and topology)

Chicago, IL

Invited student

Summers 2015, 2016

TEACHING

The Institute for Emerging CORE Methods in Data Science

La Jolla, CA

Lecturer for the Foundations in Data Science high school summer program

Jul 2023—Aug 2023

- Designed and presented 5 lectures/homework for a module on linear algebra; director: *Rajiv Gandhi*

University of California, San Diego

La Jolla, CA

Teaching assistant

- Machine Learning, *Yian Ma*, Fall 2022; Probability and Statistics, *Sanjoy Dasgupta*, Fall 2020

Columbia University

New York, NY

Teaching assistant

- Awarded fellowship; presented 15 hours of lectures on unsupervised learning techniques; designed 9 homework problems; taught over 100 hours during office hours and individual meetings
- Unsupervised Learning, *Nakul Verma*, Summer 2018; Machine Learning, *Nakul Verma*, Summer 2018; Graph Theory, *Tim Sun*, Spring 2018; Geographic Information Systems, *Michael Parrott*, Fall 2017

SKILLS

Python, Pytorch, C, SQL

AWARDS

2020 UC San Diego Changemaker Challenge 1st Place

In the UC San Diego COVID-19 Contact Tracing Challenge, sponsored by XYO

2019 Andrew P. Kosoresow Memorial Award for Excellence in Teaching and Service

Awarded for outstanding contributions to teaching in the Department of Computer Science at Columbia University and exemplary service to the Department and its mission

SERVICE

Reviewing

ACM CSUR 2024; AISTATS 2022–2025; ALT 2025; ICML 2025; JOTA 2025; NeurIPS 2023, 2025

EnCORE Outreach Volunteer

La Jolla, CA

Volunteer instructor

UC San Diego Diversity Fellowship Committee

Reviewer

La Jolla, CA

*Jan 2020, Jan 2021***CSE/HDSI Ph.D. Visit Day**

Coordinator for AI/ML group

La Jolla, CA

*Mar 2020, Mar 2021***Google/UC San Diego ExploreCSR Mentorship Program**

Volunteer mentor

San Diego, CA

Oct 2019—Jun 2020

- Designed/taught a computational thinking course for underserved students in computer science

Friends of Washington Park

Volunteer mentor

Chicago, IL

Jan 2014—Jul 2017

- Tutored 5th–8th grader students in an after-school program in local neighborhood of Hyde Park

TALKS**Paper talks and posters**

- Symposium on Mathematical Foundations of Trustworthy Learning (ETH Zurich) *Oct 2025*
- Chicago Junior Theorists Workshop (Northwestern and TTIC) *Dec 2024*
- Simons Institute for the Theory of Computing (Berkeley) *Oct 2024*
- Signals, Information, and Algorithms Lab (MIT) *Mar 2024*
- Yale Theory Student Seminar (Yale) *Mar 2024*
- EnCORE Student Social (UCSD) *Feb 2024*
- Redwood Center for Theoretical Neuroscience (Berkeley) *Dec 2023*
- FODSI CCSI Student Posters (MIT) *Jun 2023*
- EnCORE Student Social (UCSD) *Mar 2023*

Reading seminars

- Forecasting and calibration (UCSD) *2023*
- Algorithmic game theory (UCSD) *2022—2023*
- Weak supervision for text data (Seekr) *2022*
- Unsupervised learning (UCSD) *2022*
- Sampling/MCMC (UCSD) *2021—2022*
- Neural networks (UCSD) *2020*
- Learning theory (UCSD) *2019*
- Sums-of-squares optimization (Columbia) *2018*
- Generalization theory (Columbia) *2018*
- Type theory (UChicago) *2016—2017*

Reading seminar presentations

- [Statistical and online learning: a tutorial](#) *Dec 2023*
- [Calibrated learning and games](#) *Dec 2023*
- Universal prediction of individual sequences *Nov 2023*
- [Introduction to neural nets](#) *Jul 2023*
- Convergent message passing *Jul 2023*
- Gradient-based multiobjective optimization *Apr 2023*
- Introduction to mean-field games *Feb 2023*
- [The double descent phenomenon](#) *Nov 2022*
- Multicriteria decision making *Oct 2022*
- [Learning with multi-modal data: canonical correlation analysis](#) *Sep 2022*
- Gradient-based learning in games *Apr 2022*
- [Linear system identification with reverse experience replay](#) *Apr 2022*

• Introduction to algorithmic game theory	Mar 2022
• Independent component analysis	Feb 2022
• Equilibrium computation: motivation and problems	Feb 2022
• Scalable sampling for discrete distributions	Nov 2021
• Graphical games	Nov 2021
• Active learning for maximum likelihood estimation	Oct 2021
• Stochastic calculus on manifolds: part 1, part 2	Aug 2021
• Linear system identification without mixing	Jun 2021
• Sequential kernel herding	Jun 2021
• Log-sobolev inequalities and concentration	Apr 2021
• Learning language games through interaction	Apr 2021
• Global non-convex optimization with discretized diffusion	Apr 2021
• Model of conserved macroscopic dynamics predicts future motor commands	Feb 2021
• A theory of universal learning	Nov 2020
• Oja's rule for streaming PCA	Sep 2020
• Proving the lottery ticket hypothesis	Aug 2020
• Approximate guarantees for dictionary learning	Jun 2020
• k -SVD for dictionary learning	May 2020
• Proximal methods for hierarchical sparse coding	May 2020
• Transformers are universal approximators	Apr 2020
• Using SVD to learn HMMs	Feb 2020
• Conditional mutual information and generalization	Feb 2020
• Generalization and adaptive data analysis	Jan 2020
• Generalization and differential privacy	Nov 2019
• Invariant risk minimization	Nov 2019
• Complexity: beyond space and time	Aug 2019
• Zero-knowledge proofs from MPCs	Apr 2019
• Geometry of gradient descent and lower bounds	Feb 2019
• Homomorphic encryption	Feb 2019
• Approximate nearest-neighbor search	Dec 2018
• Introduction to tensor decompositions	Dec 2018
• Sums-of-squares for robust estimation	Nov 2018
• Spectral graph theory, earlier version	Oct 2018
• Sums-of-squares for maxcut	Sep 2018
• Tensor decomposition for parametric estimation	Jul 2018
• PAC-Bayes for neural networks	Apr 2018